# CHAPTER 1

# Understanding the Stock Market

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The stock market is crucial in modern financial markets. It helps generate wealth, reflects economic health, and drives growth. Companies use it to raise capital for expansion and innovation. Savers and investors participate in the market to grow their wealth, and it also serves as a direct expression of various economic indicators.

Stock markets, which have existed since the 1600s, started as physical locations where traders would meet to buy and sell shares but stock markets have since transitioned into digital platforms that operate worldwide. In the stock market, buyers and sellers trade stocks, representing ownership in public companies. These transactions involve much more than a simple money exchange for ownership rights. They combine economic indicators, investor sentiment, and global events. A high valuation of a company's stock often shows a vote of confidence from the markets. Conversely, declining stock prices can signify concerns about a company's prospects or broader economic issues.

The stock market impacts retirement portfolios, college savings plans, and even the economic policies of governments. Its movements can influence consumer and business confidence and shape spending and investment decisions on a global scale. Therefore, understanding the stock market goes beyond comprehending the rise and fall of individual stocks. It involves grasping the relationship between these movements and broader economic narratives and recognizing their impact.

This chapter will teach you about the stock market's core concepts. If you understand the stock market basics, learning the specifics of implementing trading strategies in future chapters will be easier. If you already possess extensive stock market knowledge, feel free to skim this chapter. However, if you are new to investing, reading and comprehending each concept is valuable.

We will start by briefly exploring the stock market's function and history. Next, we will examine different financial markets, each with a distinct role in the global financial ecosystem and some investment vehicles. You will then learn core concepts in stock trading, from what stocks (also called shares) are to the diverse market participants and their roles. This section will help you comprehend how stock markets function daily.

Analyzing stocks is historically more of an art than a science, and we will study the details and limitations of fundamental and technical analysis, two popular methodologies investors apply to select investment opportunities. Understanding how these approaches work is essential to engaging in the stock market because many participants use them, even if they have drawbacks.

Afterward, we will learn about portfolio management and trading strategies and discuss how to build and manage a diversified investment portfolio. Moreover, we will overview various investment approaches, from passive investment to value and growth investing.

By the end of this chapter, you will have a solid foundation in stock market basics. A solid understanding of these principles is necessary for using artificial intelligence (AI) in financial forecasting. As technologies like AI become more integrated with financial decision-making, having such knowledge becomes indispensable.

## Stock Market Fundamentals

Investors actively buy and sell company shares in the stock market. The marketplace not only facilitates stock trading but also serves as a platform for companies to raise capital from the public. This capital is critical for companies to fund operations, innovate, and expand, thus making it vital to economic growth and development.

The stock market's significance extends beyond the corporate sphere. For investors, it represents a means to grow wealth and save to

achieve long-term goals like retirement or education. For the economy, it acts as an indicator of health and confidence. Robust and active stock markets often signal a thriving economy, while downturns can show economic challenges. Policymakers, economists, and investors monitor the stock market's health, as it can offer significant insights into the economy.

## HOW STOCK MARKETS CONTRIBUTE TO ECONOMIES

Stock markets fulfill several vital economic functions. First, they provide a mechanism for price discovery, where supply and demand meet to price publicly traded companies. This pricing mechanism is important as it reflects the market's collective judgment of a company's worth, based on its current performance and prospects.

Second, stock markets offer liquidity, allowing investors to buy or sell shares cheaply. This liquidity makes investing in stocks attractive and enables companies to raise capital more efficiently. When a company issues shares, it can tap into a broad pool of potential investors, often more helpful than seeking funding through loans.

Third, stock markets facilitate capital allocation. Efficient markets ensure that investment toward companies and industries performing well uses funds in the most productive way possible and has growth potential. This allocation plays a significant role in driving economic innovation and growth.

Finally, stock markets serve as a key barometer for the economy, reflecting investor sentiment and expectations about the future. Movements in stock prices can provide insights into how investors view the economy's health, influencing consumer confidence and business decisions.

The stock market's role in price discovery, providing liquidity, allocating capital, and serving as an economic indicator underscores its importance to investors, companies, and the broader economy. In addition to the stock market, other related financial markets support the financial ecosystem.

## DIFFERENT FINANCIAL MARKETS

The financial market is a broad term encompassing various marketplaces where participants can trade financial assets, commodities, and other fungible value items at prices determined by supply and demand.

While the stock market is the most recognized form of a financial market, others exist, and understanding them is important.

These markets, which vary in scale, function, and the types of assets traded, collectively facilitate the efficient movement of capital and risk throughout the economy. They serve as venues for raising money, investing, risk management, and price discovery. Each market type has unique characteristics, rules, and participant dynamics, and each of these markets plays a specific role in the financial system:

- Stock Markets: They deal with trading shares, which are ownership units of companies. These markets are pivotal for companies seeking to raise capital and investors looking to buy partial ownership of these companies. Investors and policymakers often see stock markets as indicators of economic health and business prosperity.
- Bond Markets: Unlike stock markets, bond markets trade debt securities. This market allows governments, municipalities, and corporations to raise funds for various projects or operations. Investors (lenders) in bond markets provide money to the bond issuer (the borrower). In return, investors receive interest payments over time, plus the bond's total value when it comes due. Bond markets serve as interest rate and credit risk indicators.
- Commodities Markets: These markets involve trading physical goods or primary products such as gold, oil, and agricultural products. Commodities markets are vital for managing supply and demand dynamics across different industries, allowing producers and consumers to hedge against price volatility.
- Forex (Foreign Exchange) Markets: Investors trade currencies in the forex markets 24 hours a day, making it one of the most liquid markets. Forex markets are essential for global trade and finance integration, as they allow for exchanging different currencies, facilitating international business and investment.

Other complex financial instruments, like futures and options, are traded in derivatives markets. These instruments derive their value from the value of an underlying asset. The underlying assets usually include stocks, bonds, commodities, or currencies.

Futures are contracts allowing the buying or selling of an asset in the future at an agreed price. Exchanges standardize futures contracts in quantity and quality to simplify their trading. Investors employ futures for risk management or speculation. Options are instruments that give the holder the right to buy or sell an asset at a prespecified price before a specific date. But unlike futures, they do not require buying or selling the asset. Call options give the right to buy, and put options give the right to sell.

Participants can use derivatives as powerful tools for managing financial risk and speculative purposes. However, they can also be complex and carry high risk. Hence, it is imperative to comprehend these derivative instruments before trading them.

The financial market landscape is diverse, with each market type serving distinct purposes within the global economy. From providing avenues for raising capital (stock and bond markets) to facilitating international trade (forex market) and offering platforms for hedging against price volatility (commodities and derivatives markets), these markets are integral to the financial system. In Part II of the book, you will learn how these diverse markets present challenges and opportunities for AI-driven analysis and decision-making.

## INVESTMENT VEHICLES IN THE STOCK MARKET

Before learning about the fundamental stock trading concepts, we must understand how investors can participate in the stock market. Different investment vehicles offer diverse levels of risk, management styles, and investment strategies.

- **Mutual Funds:** They gather money from many investors to create a managed portfolio of diverse stocks, bonds, or other securities. Investors buy shares in the fund to benefit from diversification and professional expertise. The fund regularly divides the total resources among various investments, like stocks or bonds. These instruments are ideal for those looking for diversified portfolios and professional management.
- **Exchange-Traded Funds (ETFs):** These instruments are investment funds traded on stock exchanges. ETFs track market indexes, commodities, or bundles of assets. They offer seamless flexibility, allowing investors to exchange them at market prices and low costs throughout the day.

- **Self-Managed Accounts:** For those who prefer a hands-on approach, self-managed accounts allow individuals to buy and sell stocks, bonds, and other securities according to their research and strategies. This method requires more knowledge and time investment but offers complete control over the trading decisions.
- **Robo-Advisors:** These automated platforms effortlessly create and manage a diversified portfolio. They use algorithms to select investments based on your risk tolerance and goals, making them an option for those seeking hands-off investment management.
- **Hedge Funds:** These are alternative investment funds that can pursue more complex trading strategies, like using leverage, trading derivatives, and short selling. Hedge funds typically require a high net worth to invest in them, but theoretically they can provide returns independent of overall market conditions.

Active managers of some mutual and hedge funds and investors with self-managed accounts typically engage in active stock trading.

## Stock Trading

Stock trading is buying or selling company shares to generate a profit. It combines economic insights, market analysis, and a psychological understanding of the markets. While trading stocks might seem straightforward, the process involves understanding several fundamental principles. These principles include the mechanics of trading, the stocks available, the roles of various market participants, and the strategies deployed in trading decisions.

### WHAT ARE STOCKS?

Stocks represent part ownership in a company. The term "share" is often used interchangeably with stocks, although it usually represents the units of stocks of a specific company. Individuals who buy a company's stock become shareholders. Investors own a fraction of a company in proportion to the total shares they purchase. For example, if a company has 100 million shares and you buy 1 million, you own 1% of the company.

Stocks can be common or preferred.

- Common Stocks: They are the most usual type of stock. They give shareholders voting rights, usually one vote per share owned. Shareholders sometimes receive dividends and can vote to choose company board members who oversee management decisions. Stock markets typically trade common stocks.
- Preferred Stocks: They rarely have voting rights but have a higher claim to company assets and earnings than common stocks. Preferred shareholders get dividend payments before common shareholders. In the event of company liquidation, they also receive payment before others.

Investors sometimes categorize companies by their market capitalization, which is the total market value of their outstanding shares. This classification helps investors understand the company's size and includes categories like large-cap, mid-cap, and small-cap. Small-cap shares have higher trading costs. Because of these high trading costs, many active portfolio managers avoid trading them. Since fewer market participants actively incorporate the latest information in them by trading, their returns are usually easier to predict. Because of their predictability, they are an excellent choice for investors, with low fees and minimal market impact.

## MARKET PARTICIPANTS

The stock market has various participants, each playing a unique role. One of the key groups is individual investors, also called retail investors. These are everyday people who buy and sell stocks. They have distinct objectives, from building a portfolio for long-term goals like retirement or education savings to actively trading, hoping to achieve short-term gains.

Another significant group is institutional investors. This category includes pension funds, mutual funds, hedge funds, and insurance companies that invest sizeable sums of money. They can substantially influence market prices because of the size of their trades.

Brokers and dealers also play an important part. Brokers facilitate the buying and selling of stocks for clients, while dealers trade

stocks for their accounts. Moreover, market makers provide liquidity by buying and selling stocks to ensure organized trading.

Lastly, regulators and central banks form an essential component of the market. Institutions such as the Securities and Exchange Commission (SEC) oversee the markets. Their primary goal is to guarantee fairness in trading and to uphold transparency in all market activities. Central banks control interest rates and affect firms' valuations. Sometimes, during extreme market periods, they also buy and sell instruments directly.

## BUYING AND SELLING STOCKS

Besides the theoretical discussion, understanding the practical steps in buying and selling stocks is beneficial. Here are some potential specific steps.

1. **Opening a brokerage account:** To trade, an individual must open an account with a brokerage firm. This process involves providing personal information and answering questions about investment experience and risk tolerance.
2. **Researching:** Before investing, it is vital to study the stocks, which involves analyzing the company's financial statements, market position, and growth potential.
3. **Placing trades:** Traders can place orders to buy or sell stocks through a broker or an online trading platform. Common order types include market orders, limit orders, and stop-loss orders.

Market orders are instructions on buying or selling a financial instrument immediately at the current price. These orders ensure that the trade happens rapidly but do not guarantee the price. Second, limit orders instruct brokers to trade only at a specified price or better. Buy limit orders set the highest price to pay. Sell limit orders specify the lowest price to accept. Limit orders guarantee a price level, but not that the trade will happen.

Stop-loss orders are a more advanced type. A market stop-loss order attempts to limit an investor's loss in a trading position. Stop-loss orders set a trigger price to cut losses if a stock's price changes adversely. If the stock hits the trigger price, the order is executed as

a market order at the next available price. Once the stock hits this price, the stop-loss order automatically converts into a market order and trades at the next available price. Stop-losses cap downside risk but do not guarantee the final sell price. In fast markets, the actual price can be much lower. Traders can place stop-loss orders for buying or selling.

Trading stocks involves regularly monitoring stock performance and staying informed about market conditions. Later in the chapter, we will learn how diversification, investing across various stocks or asset classes, can help you reduce risk. Now that you have learned about these foundational concepts, from understanding what stocks are to recognizing the roles of different market participants, we will look at classic investment strategies.

## CLASSIC STOCK-PICKING STRATEGIES

Stock investment strategies involve identifying which companies to invest in. The analysis is not just about picking winners and avoiding losers; it is also a systematic approach to understanding a stock's potential risks and rewards. In this section, you will learn about two historical methods of stock research: fundamental analysis and technical analysis. Moreover, we will also cover news trading.

### FUNDAMENTAL ANALYSIS

Fundamental analysis tries to calculate a stock's intrinsic value by studying how economic, financial, and other factors affect the company's value through its business and prospects. Investors perform several types of analysis with the hope of obtaining a measure of the "true" value of a stock:

- **Economic Analysis:** It involves looking at the overall economic environment, including factors like interest rates, inflation, and economic growth, and how it can affect a company's performance.
- **Industry Analysis:** Analyzing the industry involves understanding how the market works, including competition, growth opportunities, and regulations. This analysis helps in identifying the best-performing companies within a sector.

- **Company Analysis:** This is a thorough analysis of a company's financial statements—income statement, balance sheet, and cash flow statement. Analysts look at revenue, profit margins, return on equity, and earnings per share to gauge the company's financial health.
- **Valuation Metrics:** These include the price-to-earnings (P/E) ratio, price-to-book (P/B) ratio, dividend yield, and others. They help determine whether a stock is undervalued or overvalued compared to its historical performance or sector averages.

The intuition of fundamental analysis is that sometimes the current market value of a company differs from its intrinsic or true value since investors may not always price every company correctly. This strategy tries to assess a company's fair value, but several limitations can complicate the process:

- **Intrinsic Value Subjectivity:** Fundamental analysis involves estimating the inherent value of a stock based on economic, financial, and industry data. However, this process is highly subjective. Due to varying assumptions and models, analysts may produce different valuations based on the same data.
- **Impact of Non-Quantifiable Factors:** Management quality, brand value, and market competition are challenging to quantify but they significantly affect a company's performance. Fundamental analysis that is based on firms' financials might not fully account for these qualitative aspects.
- **Delayed Reflection of Current Events:** Fundamental indicators, like earnings reports, economic data releases, and industry trends, are historical and may not reflect the latest market changes or future uncertainties. As a result, investors might react slowly to new developments.
- **Market Irrationality:** Sometimes, the market does not behave rationally, and stock prices do not reflect underlying fundamentals because of speculative activities, market sentiment, or other macroeconomic factors. While fundamental analysis hopes the prices will correct in the future, this correction may take too long, and investors may be forced to close their positions with losses.

- **Estimating Future Earnings:** While earnings per share (EPS) or price-to-earnings (P/E) ratios are staples in fundamental analysis, forecasting future earnings is uncertain. Various unpredictable factors, such as market disruptions, new competitors, regulatory changes, or global events, can influence a company's future profitability. Relying on projected earnings can sometimes lead investors astray when these projections are based on optimistic or pessimistic assumptions.
- **The Unreliability of Book Value:** The price-to-book (P/B) ratio is a fundamental analysis signal based on a company's book value. However, a company's assets minus its liabilities can sometimes mislead book value because it does not correctly quantify intangible assets such as brand value and intellectual property. Outdated asset valuations or unaccounted liabilities can skew the accurate picture. Relying on the P/B ratio can cause misjudgments about a company's intrinsic value.

### *Technical Analysis*

Unlike fundamental analysis, technical analysis assumes that historical trading activity and price changes can be reliable indicators of future stock performance. Technical analysts use various graphical displays of price volume called price charts. These include line, bar, and candlestick charts. Each type of chart depicts unique insights into market behavior. Traders try to predict future prices by examining these charts using some standard techniques, including:

- **Trend Analysis:** They identify the market's direction or a specific stock's price. Trends can be upward, downward, or sideways and are crucial to technical analysis.
- **Technical Indicators and Patterns:** Analysts use a range of indicators with distinctive names like moving averages, Relative Strength Index (RSI), and Bollinger Bands. They also look for patterns like head and shoulders, double tops and bottoms, and others that historically show potential market moves.
- **Volume Analysis:** Examining trading volume can, in theory, show if a price trend is reliable. Higher volumes validate the trend, while lower volumes may suggest a lack of conviction among traders.

- **Sentiment Analysis:** It involves assessing the overall mood of the market, which can be bullish (expecting prices to rise) or bearish (expecting prices to fall). News events, market rumors, and investor psychology can influence sentiment.

Despite the academic skepticism about technical analysis, it remains a widely employed technique. One theory is that it works because of **self-fulfilling prophecies**. Suppose many traders believe in a particular technical signal (e.g., a moving average crossover) and act upon it. In that case, their collective actions can indeed influence the stock price in the expected direction. The price change can be unrelated to the inherent validity of the signal and, instead, happen because the signal triggers a herd behavior among the traders that follow it.

Like fundamental analysis, technical analysis also has several limitations:

- **Pattern Reliability:** Technical analysis assumes that history repeats. However, the reliability of patterns and trends is often questionable, as unexpected market events or news can disrupt them, leading to false signals.
- **Overemphasis on Price Data:** Technical analysis predominantly focuses on price movements and volume, potentially overlooking broader economic or sectoral trends affecting a stock's future performance.
- **Over-Reliance on Past Data:** Technical analysis is retrospective. It assumes that historical price movements can predict future performance. However, many unpredictable factors influence stock markets, from sudden geopolitical events to surprise earnings reports. Relying only on past patterns can often leave traders ill-prepared for unexpected market shifts.

Besides theoretical limitations, technical analysis techniques face practical challenges during implementation:

- **Overwhelming Number of Indicators:** There are hundreds of indicators in technical analysis, and their number can be overwhelming. The variety of tools may lead traders to

become flooded with conflicting signals and struggle to make optimal trading decisions.

- **Subjectivity:** Two traders looking at the same chart will reach different conclusions. Subjectivity is likely to arise if the analysts have strong feelings about the firm being analyzed or feel happy or sad that day. It can lead to inconsistent predictions.
- **Lack of Fundamental Consideration:** Technical analysis, in its purest form, disregards a company's fundamentals. Price and volume trend data can provide insights into short-term market sentiment but do not reveal a company's underlying health or potential. Ignoring fundamentals can sometimes lead to missed opportunities or undue risks, especially in the long term.
- **Short-Term focus:** Many technical analysis strategies target short-term trading opportunities. This horizon can be advantageous for day traders or swing traders but may not align with investors' goals for long-term growth and value. Relying only on technical analysis can be impractical for those with extended investment horizons.

Further, several practical implementation challenges apply to fundamental and technical analysis:

- **Cost:** One of the primary drawbacks of these techniques is their cost-intensive nature. Fundamental analysis requires extensive time and money to review companies' finances, industry reports, and economic data. Technical analysis is expensive because of the need for continuous market monitoring and time investment in learning various tools. Hiring skilled analysts, purchasing detailed reports, and dedicating time to in-depth research can be very costly, especially for smaller firms with fewer resources.
- **Scalability:** These approaches are not scalable. As an investment firm expands its portfolio, the requisite research grows, creating a bottleneck effect. Studying many companies in various sectors and regions increases complexity and complicates maintaining accuracy and detail.

- **Undiversified Portfolios:** Given the intensive research process, investors relying only on fundamental or technical analysis might end with a concentrated portfolio. In-depth research can inadvertently cause overconfidence in a few stocks, raising the risk of sector or company-specific downturns. Such a strategy can lead to heightened portfolio volatility, especially if one or more of these "high conviction" stocks underperform.
- **Challenges for Non-Experts:** For retail investors or those without a background in finance, diving deep into financial statements or price charts can be intimidating. Understanding the details of financial signals and their future impact on stocks is challenging and requires much learning. Many investors lack the expertise, patience, or interest to understand financial statements or learn charting tools and indicators.

These high costs, scalability issues, and investor biases can lead to volatility and suboptimal returns. Subjective interpretations and inherent complexities limit the accuracy of predictions in market analysis. In contrast, news trading is a more straightforward trading strategy. In the later chapters, we provide a step-by-step guide to implementing a news trading strategy.

## NEWS TRADING AND SENTIMENT ANALYSIS

News trading involves buying or selling stocks based on news announcements. Significant economic reports, earnings announcements, and geopolitical developments can lead to abrupt and sizable movements in stock prices. News trading often relies on sentiment analysis, which involves evaluating the tone and context of textual data to determine investors' collective attitude or sentiment toward a particular stock or the market overall. It uses multiple data sources, including news articles, social media posts, and financial reports. Investors approach sentiment analysis using two methods: manual and automatic.

### *Manual Sentiment Analysis*

Manual sentiment analysis involves analysts reading, interpreting, and categorizing content (like news articles, tweets, or forum posts) to determine the sentiment toward a stock or the overall market.

Humans are naturally good at understanding context, sarcasm, and nuanced language that automated systems might miss. Still, this technique has its limitations. For example:

- **Scalability:** Classifying texts by hand is labor-intensive and time-consuming, making analyzing large volumes of data costly.
- **Subjectivity:** Different analysts might interpret the same information differently, leading to inconsistent results.
- **Slower Process:** The time to read and interpret can lead to delayed reactions in fast-moving markets.

*Automatic Sentiment Analysis*

Automatic sentiment analysis uses software to quantify news sentiment. It has the advantage of handling large amounts of data, and unlike humans, machines apply the same standards across all data, leading to consistent analysis. Yet, several limitations remain:

- **Contextual Misunderstanding:** News articles and reports often contain nuances, sarcasm, or context-dependent meanings that early sentiment analysis tools cannot detect.
- **Dependence on Quality Data:** The performance depends on the training data quality.
- **Complex Set-Up:** Requires expertise to set up and maintain models, and there is a need for continuous updates as language changes.

While the early automatic sentiment analysis tools have many limitations, recent advances in AI have produced models capable of dealing with these nuances. The book's later chapters explain how ChatGPT can ease some of these limitations. Lastly, both the manual and automatic approaches face practical implementation challenges:

- **Lack of Contextual Understanding:** Non-expert investors and simple algorithms may misunderstand the sentiment of a news article or announcement. For example, some investors may view a news report about a company cutting costs positively, which can increase profit margins in the short term.

However, if the cut is in research and development, and the market understands it will lead to decreased profits in the future, the news can lead to a price decline. Traditional sentiment analysis tools and non-expert investors may miss such subtle differences.

- **Overreliance:** Non-professional investors, eager to leverage sentiment analysis, often rely too heavily on these tools, treating them as definitive predictors. This overreliance can lead to overlooking other critical market indicators.
- **Misunderstanding Market Reactions:** Retail investors and classic algorithms without a deep understanding can misjudge how other investors, especially institutional ones, would react to certain news. Even if a sentiment analysis tool labeled news "positive," investors might perceive it differently.
- **Lack of Comprehensive Coverage:** The overload of news sources available means that non-expert retail investors might only focus on mainstream news. Without a systematic approach, they risk overlooking significant pieces of information.
- **Emotional Bias:** When combined with personal biases, sentiment analysis tools can amplify the potential for misguided choices. A retail investor holding a positive bias toward a particular stock might give more weight to positive sentiment analysis, overlooking other negative signals from different estimation methods and leading to confirmation bias.
- **Trading Speed:** Sentiment analysis tools and human investors may not be able to react as quickly as high-frequency algorithmic trading systems to news events. By the time an insight or trade signal is generated, algorithms may have already capitalized on the event, and prices may have shifted. This makes trading on sentiment challenging.

Fortunately, as we will cover in the following chapters, AI can complement traditional methods, offering more dynamic, comprehensive, and data-driven approaches to stock market analysis.

## Portfolio Management and Trading Strategies

Understanding the basics of portfolio management and trading strategies is fundamental to investing in the stock market. While stock

analysis helps investors decide which stocks to buy or sell, portfolio management and trading strategies determine how these decisions fit into a broader investment plan.

## BASICS OF PORTFOLIO MANAGEMENT

Portfolio management involves creating and maintaining an investment portfolio that aligns with an investor's goals, risk tolerance, and investment horizon. Effective portfolio management and trading strategies are instrumental in achieving investment goals.

One of the core principles of portfolio management is diversification, which involves spreading investments across distinct assets or asset classes to mitigate risk. Diversification reduces a portfolio's volatility, as the better performance of some investments can offset the poor performance of others. Distributing across asset classes that do not move together is even better.

In asset allocation, investors weigh different categories of assets in a portfolio. These categories include stocks, bonds, and cash. The distribution should reflect the investor's risk tolerance, investment objectives, and time horizon. Regularly reviewing and rebalancing a portfolio is key to maintaining the desired asset allocation. This process involves buying or selling assets in the portfolio to match the original or updated investment strategy.

Classical investing approaches range from active trading to passive investing. Trading strategies like value investing, growth investing, and others offer suggestions for stock selection based on individual investment preferences and risk tolerance. In contrast, passive investing involves buying and holding a diversified portfolio for a lengthy period.

## COMMON INVESTMENT APPROACHES

The most straightforward and most recommended strategy is passive investing. Passive investors invest long term, avoiding the frequent buying and selling of stocks. The most common form of passive investing involves buying index funds or ETFs that track a market index like the S&P 500. Passive investors aim to replicate market performance instead of trying to beat the market. By not trading frequently, passive investors save on transaction fees and incur lower management fees, primarily when invested in index funds or ETFs. Passive strategies often involve a diversified portfolio that mirrors a broad market index, spreading risk across various sectors and assets.

Value investing is an active strategy involving buying undervalued stocks at lower prices than their intrinsic value. Investors using this strategy believe the market overreacts to news, resulting in stock price movements that do not reflect a company's long-term fundamentals. The technique involves buying stocks at lower prices than their fair value and selling them once the price reflects their true worth. The non-trivial part involves correctly perceiving a company's future value and resisting price fluctuations until the hypothetical price convergence.

In contrast, growth investors seek companies exhibiting above-average growth. The focus is on companies expected to grow at a fast pace compared to others in the market. Often, these companies will appear overpriced with metrics like the price-to-earnings ratio. Investors expect their fundamental operations to increase and drive the stock price up. However, there is a risk that these optimistic expectations will not materialize.

Another popular strategy is income investing, which focuses on generating a steady income from investments. Investors look for securities that pay dividends or interest, including bonds, dividend-paying stocks, and real estate investment trusts (REITs).

Moreover, in contrarian investing, investors buy stocks when others are selling and sell when others are buying. They believe they can capitalize on significant market mispricings in securities caused by herd behavior. In contrast, momentum investment follows the markets and buys when everyone buys.

Investors convinced that the market overvalues a stock sometimes use short-selling strategies. Shorting or short selling comprises borrowing a stock, selling it, hoping the price will fall, repurchasing it later at a lower price, and paying back the interest. Investors put assets in collateral to borrow the stock. For example:

1. Suppose you think Stock Y's price of $100 is too high and consider the correct value is $80. You expect the market to understand that the price is too high at some point.
2. To capitalize on this overpricing, you borrow one share of stock.
   a. You must repay one share of stock Y plus interest.
3. You leave $100 of cash as collateral.
   a. You will get back your collateral when you repay the share.

4. You sell the share and receive $100 to invest elsewhere or hold.
5. The next day, you close the position.
   a. Because it is a very short horizon, we will round interest to zero. For longer horizons, it is important to consider the cost of interest.
6. If the price is below $100, you make money.
   a. You repurchase it for, say, $95.
   b. Pay it back for $105.
   c. In total, you make $110 − $95 − $10 = $5.
   d. Since you started with $100 of collateral (that you get back), you get a return of 5% = $5/$100.
7. Otherwise, you lose any difference above $100.

These strategies can have limited upside (since the price is at least zero) and unlimited downside (since the price can keep increasing). Hence, they require very sophisticated risk-management techniques.

Finally, there are some advanced strategies called long-shorts. Long-short strategies seek to profit from both rising and falling prices in the market. Long-shorts involve taking concurrent long positions (buying securities expected to increase in value) and short positions (short selling securities expected to decrease in value). These strategies require an initial amount of equity to leave as collateral. The goal is for the long positions to outperform the shorts, providing an overall net gain regardless of broader market direction.

A portfolio with zero cost combines matched long and short positions in correlated securities to eliminate market exposure and risk. By weighting the long and short positions, the portfolio ensures that their price movements counterbalance each other. This strategy leads to a portfolio with no net market exposure and no inherent risk based on broader market fluctuations. The goal is to capture profits from relative mispricings between the securities.

Having covered some basic investment approaches, we will now transition to how to evaluate whether a strategy is successful.

## PORTFOLIO PERFORMANCE METRICS

Before delving into how to measure investment performance, we need to understand some definitions. This material is usually covered

in the first part of a semester-long college course. Still, we will go over a summarized version:

- **Return (percentage):** The percentage profit of the investment over a period. Returns are equal to the final value, including dividends and interest minus the initial value as a percentage of the initial value. Return = (Final Value − Initial Value)/Initial Value. For example, if you buy a stock at $100, receive a dividend during the period of $5 (and do not reinvest it), and sell at $120, the return is ($120 + $5 − $100)/$100 = 25/100 = 25%.
- **Risk-free rate:** The rate of return of investing in a safe asset (that always pays the same amount). The rate at which the U.S.A. borrows money is used to calculate the risk-free rate.
- **Excess return:** The return minus the risk-free rate.
- **Beta:** It measures the systematic volatility of an investment or portfolio compared to the market. Investors calculate it by comparing the covariance of the investment's returns with the market's returns divided by the market variance.
  - Beta = Cov(return, market return)/Var(market return)
  - A beta greater than one implies that the investment has more extreme movements than the market in the same direction as the market.
- **The Capital Asset Pricing Model** (**CAPM)** is a theory that predicts that a stock's expected excess return equals its beta times the expected excess return of the market. E[excess return] = beta*E[market's excess return].
- **Alpha** is the difference between an investment's average excess returns and the CAPM prediction. A positive alpha means the investment has performed better than its beta would predict, and vice versa.
- **Volatility**, the return's standard deviation, measures riskiness.
- The **Sharpe ratio,** an investment's average excess returns divided by its standard deviation, quantifies the tradeoff between risk and reward.
- The **Sortino ratio** is the average excess returns divided by the semi-deviation. It serves as another measure of risk, focusing only on the downside volatility.

- **Maximum drawdown (MDD)** quantifies the largest percentage drop from the highest portfolio value (peak) to the lowest value (trough) before achieving a new height. It calculates the maximum loss an investor would have experienced during a specific time frame. For example, if the portfolio starts at $50, doubles to $100, falls to $80, and recovers to $110, the maximum drawdown is (100 − 80)/100 = 20%.

Investors typically assess portfolio performance by combining these metrics. Looking at the total performance captured by the returns and the risk-adjusted performance is crucial. The alpha generally measures this performance, which assesses how the investment performed after accounting for market risk. The volatility and the maximum drawdown ordinarily measure risk. You want high returns with high alpha and low risk. Performance metrics will become helpful in later chapters when we implement a strategy step by step.

While this chapter may seem a little abstract, comprehending the fundamentals of the stock market sets a robust foundation for understanding AI's transformative role in finance. ChatGPT is opening new investment opportunities, but it is imperative to learn the basics first to exploit them. The effectiveness of these new technologies is grounded in a deep understanding of stock market fundamentals.

## Summary

In this chapter, we worked on understanding the fundamentals of the stock market. First, we explored the vital functions of the stock market in enabling trading, raising capital, allocating resources efficiently, and indicating economic health. You learned about related financial markets like bonds, commodities, forex, and derivatives, which also play critical economic roles. We covered how stocks represent ownership and how vehicles like mutual funds, ETFs, self-managed accounts, and robo-advisors allow market participation.

Next, we studied the fundamentals of trading, which involve buying and selling stocks through a broker for profit. You learned how stocks are classified by market capitalization and differ as common (with voting rights) or preferred (with higher dividends). We overviewed key market participants like investors, brokers, dealers, and

regulators that enable organized trading. We also learned about the different types of orders, like limit, market, and stop-loss.

Then, we explored popular techniques like fundamental analysis to estimate intrinsic value and technical analysis to study price charts. We covered how both face limitations like subjectivity. We also learned about news trading, which leverages sentiment analysis of announcements, either manually or using algorithms, to make timely decisions.

After that, we dived into effective portfolio management and how to align investments to goals through diversification and asset allocation while balancing risk. You learned about common long-term strategies like passive investing, value, growth, and income investing. We also covered contrarian approaches like short-selling and long-shorts.

Finally, we studied some terms that help us understand portfolio performance: returns, alpha, beta, volatility, and Sharpe ratio. We explored how returns quantify profits, alpha measures excess returns, beta captures market risk, volatility calculates investment risk, and the Sharpe ratio evaluates risk-adjusted returns.

## Coming Up Next

The upcoming chapter, "Understanding Artificial Intelligence," examines the heart of AI. It covers fundamental definitions and the mechanisms that allow machines to learn, adapt, and outperform human capabilities in specific tasks. We explore the progression from traditional machine learning to the more sophisticated deep learning, which underpins many of today's financial forecasting tools.

## Check Your Understanding Questions

- Stock Market Basics:
  - What is the stock market's primary function, and how does it contribute to the economy?
  - Differentiate between common and preferred stocks. What are the key differences in terms of rights and benefits for shareholders?

- Financial Market Types:
  - Compare and contrast the following financial markets: stock, bond, commodities, and forex. What unique characteristic does each market possess?
  - Explain the concept of a derivatives market. How do derivatives like futures and options function?
- Trading Principles:
  - Describe the role of market makers in the stock market. Why are they essential for maintaining market liquidity?
  - What is a dividend, and how does it influence the investment appeal of a stock?
- Stock Analysis Techniques:
  - Fundamental vs. Technical Analysis: Outline the primary focus of each analysis method and give an example of how each is applied in stock trading.
  - Why is volume critical in technical analysis? What information does it provide to traders?
- Portfolio Management:
  - What is diversification in portfolio management, and why is it considered an essential strategy for risk management?
  - Describe the concept of asset allocation. How does it differ for investors with varying risk tolerances and investment horizons?
- Trading Strategies:
  - Define value investing and growth investing. What are the primary criteria used in each strategy to select stocks?
  - What is passive investing, and how does it differ from other trading strategies regarding goals and methods?